GUIDE 2 OF 3

Which AI Tool for Which Job

A map of AI tools for students of mathematics, grouped by what you need to do.

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Pick the tool for the job, not the other way round. A chat assistant is good at explaining and bad at arithmetic; a computer algebra system is the opposite. Facts below come from each tool's official pages and were last checked in November 2026. These tools change quickly, so check the page before you rely on a feature.

Explaining and tutoring: general assistants

ChatGPT chatgpt.com	General assistant from OpenAI. Study mode asks questions and gives hints instead of the final answer.
Gemini gemini.google.com	General assistant from Google. Guided Learning walks you through a problem step by step.
Claude claude.com	General assistant from Anthropic.

Watch out: They write wrong answers as fluently as right ones. Age rules differ: ChatGPT is 13+ with a parent's permission under 18, Gemini needs a Google Account you may manage yourself (14+ in Italy), and Claude is 18+.

Studying from your own material

Gemini Notebook (formerly NotebookLM) notebook.google	Answers questions about the notes, slides, and PDFs you upload, cites the passages it used, and makes study guides, flashcards, and quizzes.

Watch out: Answers come only from what you upload, so a gap in your notes is a gap in the answer. Open the cited passage before you trust a summary.

Computing and checking

Wolfram|Alpha wolframalpha.com	Solves equations, limits, and integrals from plain-language input. Step-by-step solutions need Pro.
GeoGebra CAS Calculator geogebra.org/cas	Free computer algebra in the browser: solve, factor, differentiate.
Desmos Graphing Calculator desmos.com/calculator	Plot a function to check zeros, signs, and asymptotes.

Watch out: These compute exactly, but only what you typed. Check that the input is the problem you meant.

University work: LaTeX, code, papers, proofs

Overleaf AI features docs.overleaf.com	Error Assist explains LaTeX errors and suggests fixes. Free within daily limits.
Gemini in Colab developers.google.com/colab	Python notebooks where Gemini generates, explains, and debugs code. Learn Mode guides you instead of writing the code.
Semantic Scholar semanticscholar.org	Search research papers. TLDRs give a one-sentence AI summary of each result.
Lean and Mathlib lean-lang.org	Advanced: a proof assistant that checks every step of a proof, and its library of formalized mathematics (mathlib.org).

Watch out: Read every line of generated LaTeX or code before you use it. A one-sentence summary is not the paper.

Agents and personal assistants

Agents do not just answer: they carry out multi-step tasks such as browsing, running code, organising files, and scheduling.

Grok Bot x.ai/bot	Agents with their own cloud computer that use apps and websites, write and run code, and work on a schedule.
ChatGPT Work chatgpt.com/work	ChatGPT's agent for longer, multi-step tasks that end in a document, spreadsheet, or presentation.
Claude Cowork support.claude.com	Claude carries out multi-step tasks across your files and connected apps. Paid plans.
Tasks in Gemini support.google.com/gemini	Gemini runs tasks and schedules across Gmail, Calendar, Drive, and websites. 18+, paid plan, not yet available in the EU.

Watch out: Agents act on your accounts and can be confidently wrong. Review what they do, keep personal data and school accounts out, and never let them submit graded work.

Quick map

I want to …	Reach for
understand an idea or get a hint	a general assistant in study or tutor mode
revise from my own notes	Gemini Notebook
check an answer or a calculation	Wolfram|Alpha, GeoGebra CAS, Desmos
fix a LaTeX error	Overleaf Error Assist
understand or debug Python	Gemini in Colab
find papers on a topic	Semantic Scholar, then read the papers
be certain a proof is correct	Lean, or a careful human reader
hand off a multi-step chore	an agent, and check everything it did